Nuclear Reliability: Executive Summary

Vassu Doomra

Introduction

Data centers need large amounts of power around the clock, and buyers care about how much of it each reactor will actually deliver. Natural gas is the largest source of US electricity, at about 41% of generation in 2025, but it emits carbon and exposes buyers to fuel prices. Nuclear, at about 18%, supplies steady power at scale without carbon emissions which is why tech companies have signed long-term deals with specific plants:

Company	Plant	Term	Capacity (MW)
Amazon	Susquehanna, Pennsylvania	Through 2042	1,920
Meta	Clinton, Illinois	20 years from 2027	1,121
Microsoft	Crane (restart of Three Mile Island Unit 1), Pennsylvania	20 years	835

Reactors, however, do not run at full power all the time. They go offline for refueling, maintenance and unexpected faults, and the missing power must then come from somewhere else.

We asked two questions:

1. How reliably does each US reactor deliver power?
2. Can we forecast how much average power it will produce over the next three months?

Our intended user is a data center buyer or energy planner who depends on a specific reactor but can only see public data. Our contribution is a consistent forecast for each reactor that requires only public Nuclear Regulatory Commission (NRC) reports.

Deal details are from the company announcements: Talen and Amazon (June 2025), Constellation and Meta (June 2025) and Constellation and Microsoft (September 2024).

Data

We used the NRC's daily reports of power output for all 94 operating US commercial reactors, covering 2000–2025.

Source	What it provides
NRC reactor facility finder	Name, NRC region and state of each operating reactor
NRC power reactor status reports	Daily power as a % of full power, with the reason for any day below 100%

Outages. We grouped each run of consecutive zero-power days into one outage. We then used the NRC's reason text to label it as refueling, planned maintenance or unplanned outage. This gave 3,732 outages and 72,430 lost days over 25 years.

Forecast target. We averaged daily power by quarter. The target is each reactor's average power, as a % of full power, in the next quarter.

Exclusions. Vogtle 3 and 4 entered commercial operation in 2023 and 2024, too recently to have training data, so forecasts cover 92 reactors. We also removed Watts Bar 2's commissioning period before October 2016.

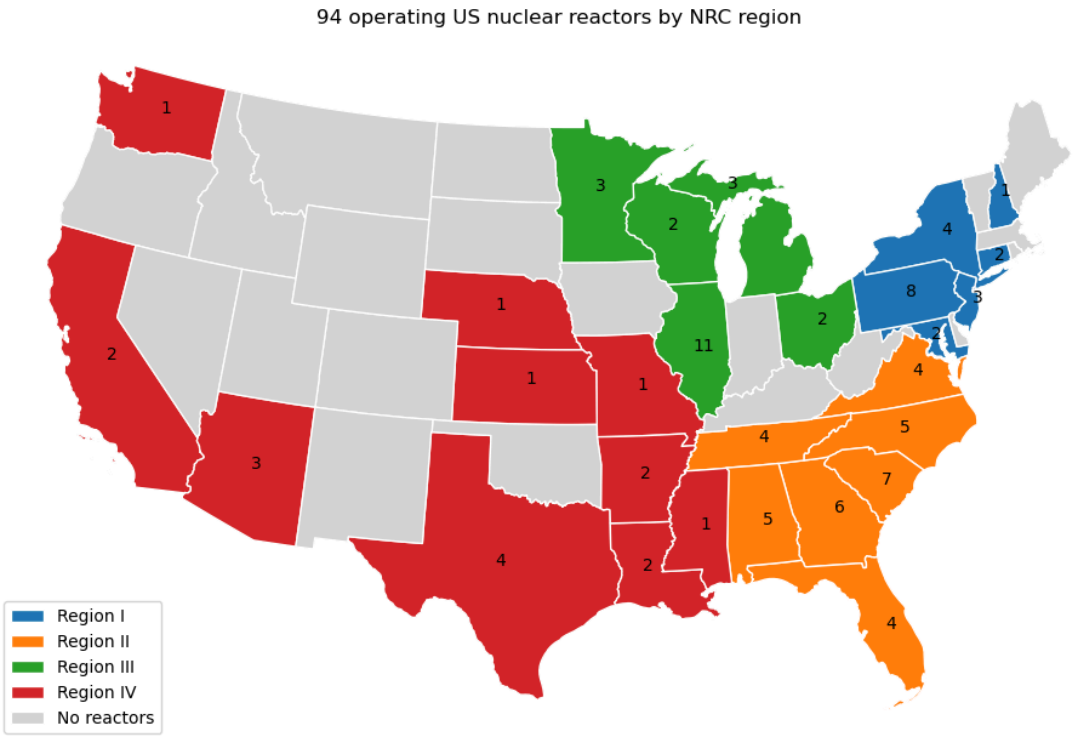


Figure 1. The 94 operating reactors by NRC region. Each colored state has at least one reactor, and the number shows how many. Color marks the NRC region: Region I has 20 reactors, Region II has 35, Region III has 21 and Region IV has 18. Illinois has the most, with 11, and states in gray have none.

Exploratory findings

The lost power is mostly driven by the refueling outages: it causes about 80% of lost days, even though unplanned shutdowns happen more often.

Outage type	Outages	Share of outages	Median length (days)	Lost days	Share of lost days
Unplanned	1,644	44.05%	3	10,121	13.97%
Refueling	1,456	39.01%	33	57,929	79.98%
Planned maintenance	574	15.38%	5	4,008	5.53%

Weather and unknown causes make up the remaining 58 outages.

Most reactors run close to 100%, with sharp drops every 18 to 24 months for refueling. There is no long-term trend, so each reactor's refueling cycle is the main pattern.

That cycle is regular but not exactly periodic, which is what makes forecasting hard:

- **Cycle lengths drift.** A refueling expected in one quarter can fall in the next.
- **Outage lengths vary.** Some refuelings take a few weeks, while others run much longer when repairs are added.
- **Unplanned outages are random.** They leave no warning in the reactor's history.

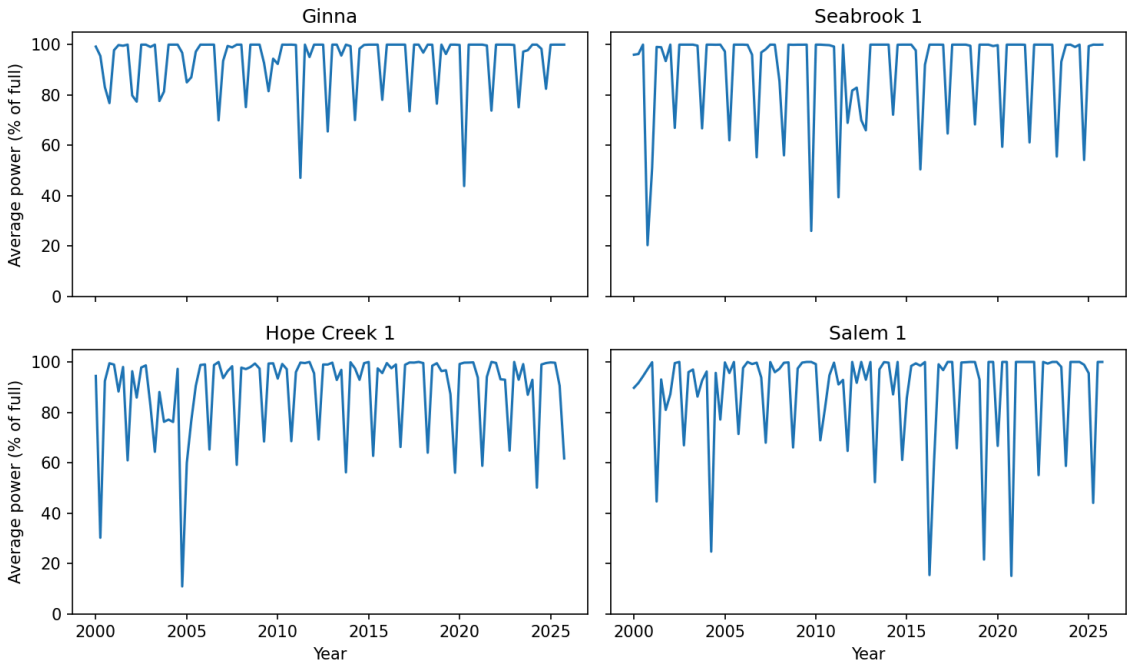


Figure 2. Quarterly average power for four Region I reactors, 2000–2025. Each panel is one reactor: Ginna, Seabrook 1, Hope Creek 1 and Salem 1. Power stays near 100%, with sharp drops that recur roughly on the refueling cycle with contributions from planned and unplanned outages as well.

How reliable is each reactor?

Which reactor a buyer signs with matters: average output over 2016–2025 ranged from 74.8% to 95.9% of full power, a gap of about 21 points.

We averaged daily power for each reactor over the last decade, so that old problems do not count against plants as they run today. This covers 92 reactors. It leaves out Vogtle 3 and 4, which started too recently, and Watts Bar 2's commissioning period.

The typical reactor is highly reliable. The median is 91.9%, and 69 of 92 reactors averaged at least 90%. A small group sits well below that: six reactors averaged under 85%. On a 1,000 MW reactor, the 21-point gap between the best and worst records equals about 210 MW of average output.

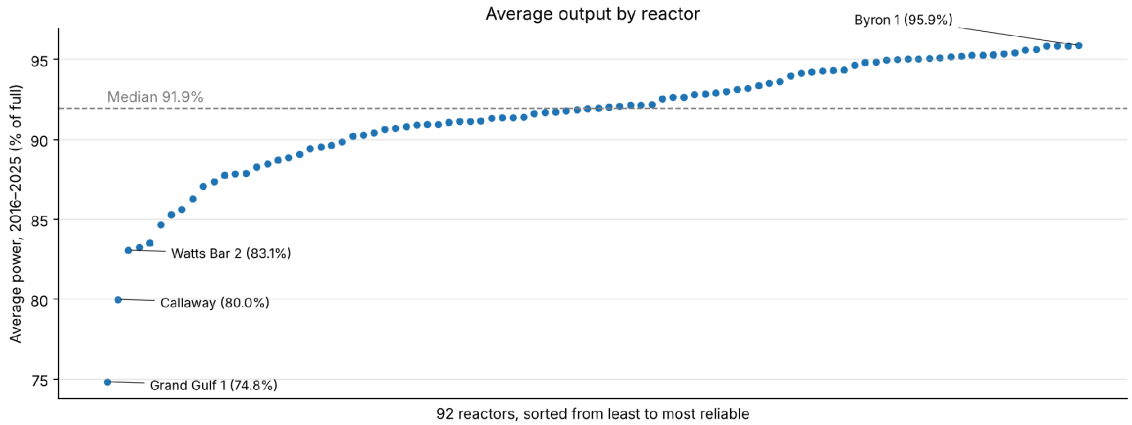


Figure 3. Average power by reactor, 2016–2025. Each dot is one reactor, sorted from lowest to highest average power. The dashed line marks the median of 91.9%. Grand Gulf 1 has the lowest average at 74.8%, and Byron 1 has the highest at 95.9%.

These ten-year averages are for reference only, and a reactor's average and ranking could change over a different time period.

Methods

We compared five models in total: two basic baselines (seasonal naive and seasonal average), two statistical models (Holt-Winters and SARIMA) and one machine learning model (XGBoost), all forecasting one quarter ahead on the same test period.

Train and test split. We trained on 2000–2022 and tested on 2023–2025, which is 12 quarters across 92 reactors. Forecasts are walk-forward, so each test quarter uses only data up to the end of the previous quarter. All tuning used the training period only.

Model	How it forecasts	What we tuned
Seasonal naive	Repeats the last refueling cycle	Nothing
Seasonal average	Averages the last *k* cycles	*k* = 2–5 over 2009–2019; chose *k* = 3
Holt-Winters	Exponential smoothing of level and seasonal pattern	α and γ per reactor, by grid search over 0.1–0.9
SARIMA	Seasonal autoregressive model	Model orders per reactor, by lowest AIC (Akaike Information Criterion) on the last 4 cycles before 2023
XGBoost	Gradient-boosted decision trees	Tree depth, learning rate and tree count, using expanding-window validation

XGBoost. The baselines only see past power. XGBoost is a gradient boosting method: it adds decision trees one at a time, and each new tree corrects the errors so far. We use the following seven features:

- Time since the last refueling, in quarters
- Length of the last refueling
- The same point in the last refueling cycle (the seasonal naive forecast)
- Average power over two days before the quarter starts
- Average power in the previous quarter
- Calendar quarter
- Reactor (Categorical Variable)

We trained XGBoost to minimize mean absolute error (MAE) and chose its settings based on the lowest validation MAE. Validation used cutoffs at 2015, 2017, 2019 and 2021, each validated through the end of 2022. The best settings were a depth of 5, a learning rate of 0.01 and 296 trees, with a validation error of 4.56, in percentage points of full power.

Metric. For each reactor we computed MAE over the 12 test quarters, in percentage points of full power. We then summarized those errors across reactors.

Results

Seasonal average was the best of the four simpler models, with a mean error of 5.59. XGBoost was the most accurate model overall: it cut mean error from 5.59 to 4.31, about 23% (95% bootstrap interval across reactors: 15% to 31%), and median error from 3.94 to 2.96.

Model	Mean	25th pct	Median	75th pct	90th pct	Max
XGBoost	**4.31**	**2.03**	**2.96**	**4.51**	**8.02**	**30.33**
Seasonal average (*k* = 3)	5.59	2.40	3.94	6.18	11.10	31.97
Seasonal naive	6.15	2.53	4.33	6.97	11.55	32.87
Holt-Winters	6.05	2.92	4.34	6.98	11.33	30.81
SARIMA	6.42	2.53	3.91	7.41	14.93	32.69

Errors per reactor in percentage points of full power, across 92 reactors, 2023–2025. Lower is better.

What this means in practice. The drop from 5.59 to 4.31 is 1.28 points of full power. On a 1,000 MW reactor, that is about 13 MW of average output, or roughly 28,000 MWh per quarter. This is an illustration that scales the average error to one reactor. It gives a sense of how much less power a buyer would have to guess at.

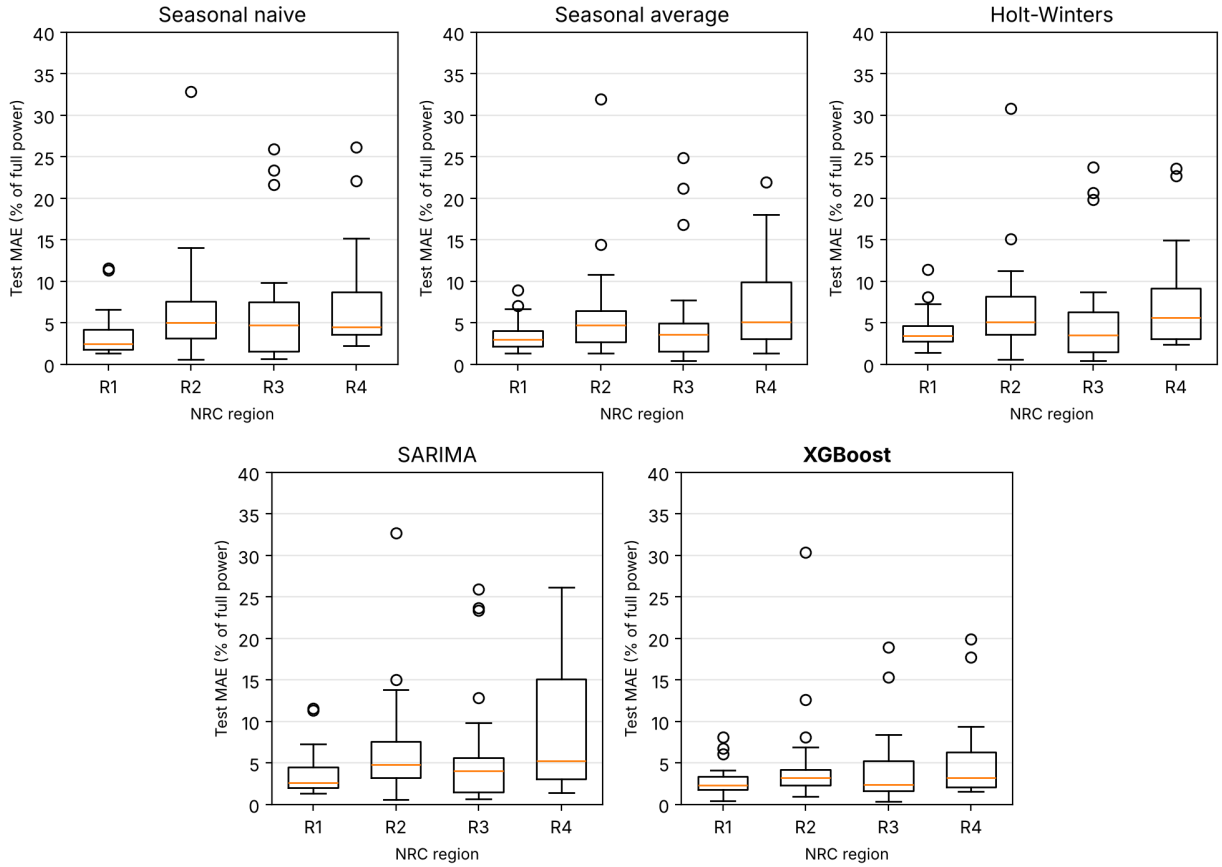


Figure 4. Test error per reactor by model and NRC region. These box plots have one panel per model, and each box shows the spread of reactor errors in one region. The orange line is the median, and lower is better. Circles are reactors that were much harder to predict. XGBoost's boxes sit lowest and are the most compact.

Where the improvement comes from

XGBoost beat the seasonal average on 70 of 92 reactors, with a median gain of 0.65 points. The largest gains came on some of the least reliable reactors: Fermi 2 improved from 24.9 to 8.4, Grand Gulf 1 from 16.1 to 4.5, and Watts Bar 2 from 10.8 to 2.4 (see Figure 3).

Watts Bar 2 shows what this looks like (Figure 5). XGBoost stayed close to actual power, while seasonal average expected dips that were too deep (see Figure 5).

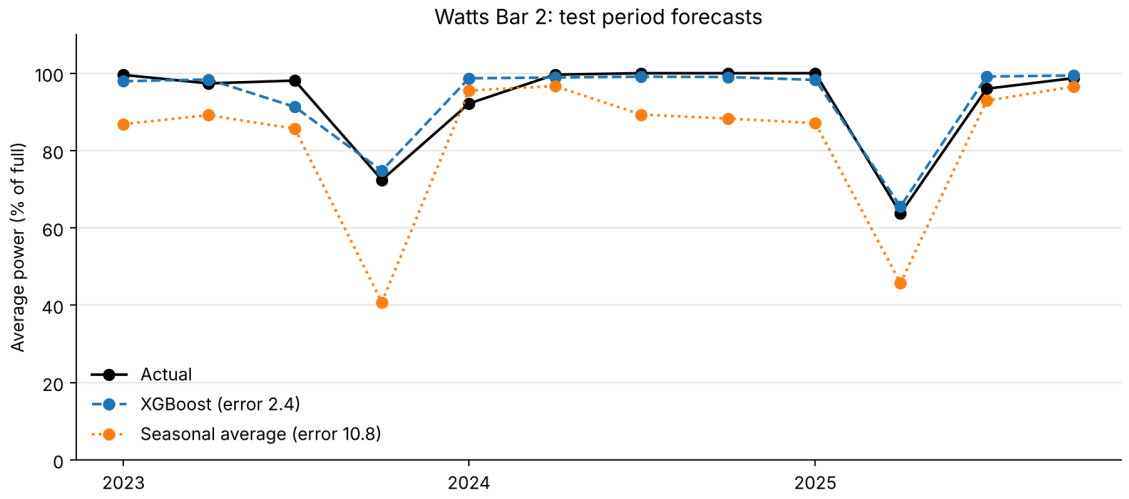


Figure 5. Forecasts against actual power for Watts Bar 2, 2023–2025. The black line is actual power. XGBoost (error 2.4) follows it closely, while seasonal average (error 10.8) runs too low in normal quarters and predicts dips that are too deep. This is one of the largest improvements, not a typical case.

For the reactors behind the Amazon and Meta deals, the gains were closer to typical: XGBoost beat the seasonal average by 1.17 points on Clinton, 0.75 on Susquehanna 1 and 0.56 on Susquehanna 2.

We call a quarter low-power if its average power was below 90% of full power. A false alarm is a forecast below 90% for a quarter that stayed at 90% or above. Most of XGBoost's improvement comes from fewer false alarms: 24, against 74 for seasonal average over our test period (across 92 reactors). As a result, its error in normal quarters was about half that of seasonal average. In the quarters that really were low-power, the two models were about tied.

Test Quarters, 2023-2025	**XGBoost**	**Seasonal Average**
False alarms (out of 861 normal quarters)	24	74
Low-power quarters flagged (out of 243)	188	183
MAE across all 861 normal quarters (% of full power)	1.54	3.04
MAE across all 243 low-power quarters (% of full power)	14.15	14.60

In short, XGBoost is better at recognizing that a quarter will be normal, but no better at predicting how deep a real shortfall will be.

What a better forecast is worth: Watts Bar 2

Let's see what a better forecast can be worth by looking at one quarter on Watts Bar 2. In Q4 2023, seasonal average expected 41% of full power, XGBoost expected 75%, and the reactor delivered 72%. We assume a capacity of 1,150 MW and a price of about $78/MWh, which is TVA's (Tennessee Valley Authority) average revenue per MWh over its 2023–2025 fiscal years.

The steps below turn that quarter's forecast errors into dollars. The quarter has 2,208 hours, and the reactor delivered 72.42% of full power.

Step	Calculation	Seasonal average	XGBoost
1. Forecast	From the model	40.78%	74.81%
2. Error in points	Difference from actual power	31.65	2.38
3. Error in MW	Points ÷ 100 × 1,150 MW	363.9	27.4
4. Error in MWh	MW × 2,208 hours	803,567	60,488
5. Error in dollars	MWh × $78	about $62.7 million	about $4.7 million

These dollar figures show the size of the forecast error, not actual money saved, and should be read with caution. Watts Bar 2 is one of the largest improvements and this is its largest gap, so this is an extreme case, not a typical one. Actual savings would depend on a buyer's contracts, backup costs and real prices at the time.

Even so, the improvement is substantial. In this quarter, XGBoost's error was about 13 times smaller than seasonal average's, and across all 12 quarters it was about 4.6 times smaller. **XGBoost also gained the most on reactors where seasonal average struggled, which makes it a strong choice for a buyer who depends on a less reliable plant.**

Why did we use $78/MWh? Watts Bar 2 is owned by the Tennessee Valley Authority (TVA), a federally owned utility that sets its own rates, so no single open-market price applies to it directly. TVA's published results give a price tied to the reactor's own operator. We divided TVA's operating revenue by its electricity sales over the three fiscal years that best match our test period.

TVA fiscal year	Operating revenue ($ million)	Electricity sales (million kWh)	Average revenue ($/MWh)	Source
2023	12,054	157,311	about 77	TVA FY2024 results
2024	12,314	162,933	about 76	TVA FY2024 results
2025	13,672	167,612	about 82	TVA FY2025 results
All three years	**38,040**	**487,856**	**about 78**	Our calculation

TVA's fiscal year runs from October to September, so these years do not line up exactly with the calendar 2023–2025. The $/MWh figures are our calculation, not numbers TVA reports. The 1,150 MW capacity comes from EIA.

Time since the last refueling was the most important feature, at 32% of total gain. The same point in the last cycle followed at 24%, and power just before the quarter at 19%. The length of the last refueling mattered least, at 4%.

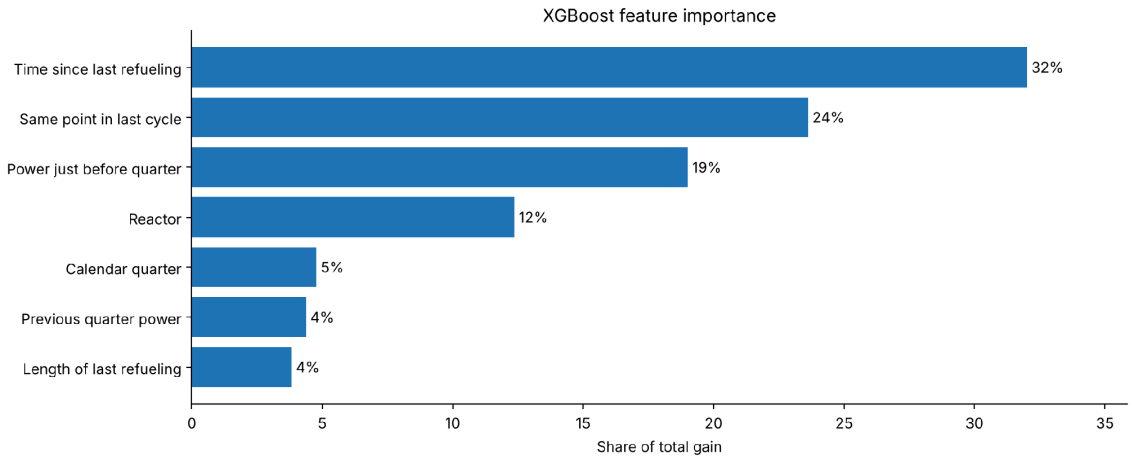


Figure 6. XGBoost feature importance. Each bar shows a feature's share of the total gain, which measures how much its splits reduced the training loss.

Where every model failed

Errors are concentrated in a few reactors. The mean sits well above the median for every model, and each model's worst error is above 30.

That worst reactor is Sequoyah 2, under XGBoost, at 30.3. Generator and turbine vibration in July 2024 led to an unplanned shutdown on July 30, which rolled into a refueling outage. The unit did not return to steady power until late June 2025. Waterford 3 (19.9) and Prairie Island 1 (19.0) were next.

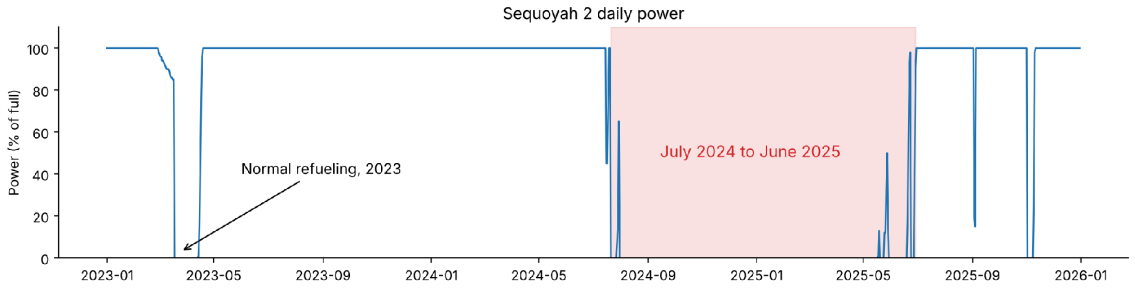


Figure 7. Sequoyah 2 daily power, 2023–2025. The dip in spring 2023 is a normal refueling outage. The shaded period marks the outage from July 2024 to June 2025, when the reactor produced almost no power. This one outage explains why every model has its largest error on this reactor.

Limitations

Rare long outages cannot be predicted from this data. The largest errors come from events that leave no trace in a reactor's power history. Sequoyah 2 ran normally until generator and turbine problems appeared in July 2024, and no feature we had could have signaled a year-long outage. Predicting such failures would need equipment-level monitoring data, which is not part of the NRC reports.

Sizing large shortfalls remains an open problem. Both models have mean errors of about 14 points in low-power quarters (including all kinds of outages), because the size and timing of these events are hard to infer from past power alone. Separate tests for refueling and unplanned outages are a natural next step.

Future outage dates are not in our data. The daily reports describe what already happened, not what utilities plan to do. Our models therefore rely only on each reactor's past refueling pattern, such as its cycle length and the time since its last refueling. When a utility moves a refueling by a few weeks, the outage can land in a different quarter than the model expects. This produces two large errors at once: one in the quarter the outage was expected, and one in the quarter it actually happened.

Outage labels depend on free-text reasons. The NRC reports explain low power in short written notes, and their wording varies across plants and years. We assigned refueling, maintenance and unplanned labels from that text, so some of the 3,732 outages may be mislabeled. Sequoyah 2 shows the difficulty: a single outage began as an unplanned shutdown and then became a refueling outage. One label cannot fully describe an outage like that, and we have no independent record to check the labels against.

We studied only reactors operating today. Reactors that shut down permanently before or during our study period are not in the data. The 92 reactors we forecast are the ones that kept running, so their record may look more reliable than the full history of the US fleet.

Next steps

Our next step is to put a dollar value on reliability for every reactor, using the same steps as the Watts Bar 2 example in the Results section (What a better forecast is worth).

```
Replacement Cost = Predicted lost MWh x Forecast Regional Electricity Price
```

A predicted power shortfall converts to megawatt-hours using the reactor's capacity. Multiplying by a regional price forecast then gives the cost of buying that power elsewhere. For a data center or a utility, this number matters: how much an outage will cost.

Summary and Conclusions

Public NRC data alone can improve next-quarter forecasts of reactor output and help a buyer judge how much to rely (or not) on a specific reactor.

Reliability differs widely between reactors. Over 2016–2025, average output ranged from 74.8% to 95.9% of full power, with a median of 91.9%. Unplanned shutdowns are the most common type of outage, but refueling caused about 80% of all lost days.

XGBoost was the most accurate of the five models we compared. Its mean error was 4.31 percentage points of full power, about 23% lower than the 5.59 of seasonal average, the best of the simpler models. It beat the seasonal average on 70 of 92 tested reactors. Most of that gain came from normal quarters, where XGBoost raised far fewer false alarms (24 against 74 over the test period). In quarters that really were low-power, the two models were about tied.

Forecasting also has clear limits. No model anticipated rare long outages such as Sequoyah 2's in 2024–2025, so a forecast can reduce routine planning errors but cannot replace a backup plan.

For a data center buyer or energy planner, this approach gives a consistent forecast of each reactor's output for the next quarter, using only public data. It is a first step toward estimating the cost of an outage before it happens.

Reproducibility

All code is in the project repository.